# Approximate Policy Iteration

Consider the infinite-horizon discounted setting with unknown model.

$$\begin{tikzcd} \pi \arrow[r, bend left, "\text{evaluation}"] & Q^\pi \arrow[l, bend left, "\text{improvement}"] \end{tikzcd}$$

Given a policy $\pi$, we want to learn $Q^\pi(s,a)$

$$Q^\pi(s,a) = \mathbb{E}_{\substack{A_t \sim \pi \\ S_{t+1} \sim P(\cdot \mid s_t, A_t)}} \left[ \sum_{t=0}^{\infty} \gamma^t R(s_t, A_t) \mid S_0 = s, A_0 = a \right]$$

using supervised learning. To create labels, we can generate sample trajectories

$$\tau^i = (s_0^i, a_0^i, r_0^i, s_1^i, a_1^i, r_1^i, \dots)$$

and set labels according to

$$y^i = \sum_{t=0}^{\infty} \gamma^t r_t^i .$$

In this case, $y^i$ will be an unbiased estimator of $Q^\pi$, i.e.,

$$\mathbb{E}_{\substack{A_t \sim \pi \\ S_{t+1} \sim P}} \left[ y^i \mid S_0 = s, a_0 = a \right] = Q^\pi(s,a) .$$

**Example:** Flipping a coin $\longrightarrow X = \begin{cases} 0 & \text{w.p. } 1-p \\ 1 & \text{w.p. } p \end{cases} \longrightarrow \mathbb{E}_X[X] = p$

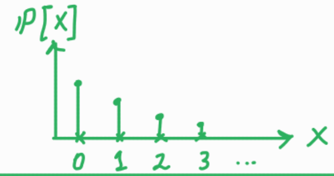
$\longrightarrow$ a sample coin flip is an unbiased estimator of the Bernoulli parameter, i.e., $\hat{p} = X^i$ and $\mathbb{E}[\hat{p}] = p$.

However, we would need an infinitely long trajectory.

Instead, we can use finite trajectories and still obtain an unbiased estimate of $Q^\pi$.

$X \sim Geo(p)$ $\rightarrow \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$

$\mathbb{P}[X = k] = (1-p)^k p \;, \; k \in \{0, 1, 2, \dots\}$




**Idea:** We use a geometric distribution with $\underbrace{\text{parameter } (1-\gamma)}_{p = 1-\gamma}$ for the length of trajectories.

$\rightarrow \mathbb{P}[X=k] = \gamma^k (1-\gamma)$

**Trajectory sampling (rollout):**

- Initialize $s_0 = s$, $a_0 = a$
- Observe $r_0 = R(s_0, a_0)$
- For $t = 0, 1, 2, \dots$ $\rightarrow t \sim Geo(1-\gamma)$
    - Terminate the run with probability $1-\gamma$; continue with probability $\gamma$ — Bernoulli trial $Ber(1-\gamma)$
    - Sample the next state $s_{t+1} \sim P(\cdot \mid s_t, a_t)$
    - Sample the next action $A_{t+1} \sim \pi(\cdot \mid s_{t+1})$
    - observe the reward $r_{t+1} = R(s_{t+1}, A_{t+1})$
- Return $\sum_{t'=0}^{t} r_{t'}$ $\rightarrow \gamma$

**Note**: With this sampling process, it holds that

$$\mathbb{E}_{\substack{A_t \sim \pi \\ S_{t+1} \sim P(\cdot \mid S_t, A_t) \\ t \sim \mathrm{Geo}(1-\gamma)}} \Big[ \underbrace{\sum_{t'=0}^{t} r_{t'}}_{y^i} \;\Big|\; s_0 = s, a_0 = a \Big] = Q^{\pi}(s_0, a_0).$$

no $\gamma^{t'}$

$\longrightarrow$

Using this rollout procedure, we can obtain samples for a state-action pair.

Now, we need a procedure to obtain samples from the entire state space and action space.

**Idea:** We sample different state-action pairs from the normalized occupancy measure, by

- generating trajectories according to $\mu_0$, $\pi$, and $P$,
- using a geometric distribution with parameter $(1-\gamma)$ for the length of trajectories.

normalized state occupancy measure:

$$\bar{\rho}^{\pi}_{s_0}(s) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s \mid s_0, \pi, P)$$

normalized state-action occupancy measure:

$$\bar{\nu}^{\pi}_{s_0}(s,a) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(S_t = s, A_t = a \mid s_0, \pi, P)$$

## Trajectory sampling (roll in):

- sample $s_0 \sim \mu_0$
- sample action $a_0 \sim \pi(\cdot \mid s_0)$
- For $t = 0, 1, 2, \dots$ $\rightarrow$ $t \sim \mathrm{Geo}(1-\gamma)$
    - Terminate the run with probability $1-\gamma$; continue with probability $\gamma$

      Bernoulli trial $\mathrm{Ber}(1-\gamma)$
    - Sample the next state $s_{t+1} \sim P(\cdot \mid s_t, a_t)$
    - sample the next action $A_{t+1} \sim \pi(\cdot \mid s_{t+1})$
- Return $(s_t, a_t)$

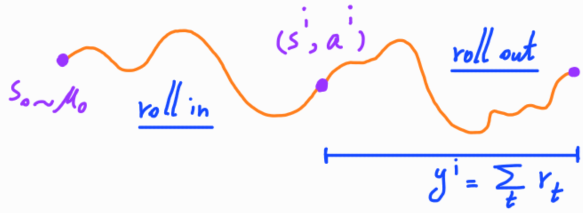
**Note**: With this sampling process, it holds that

$$\mathbb{P}\left(S_t = s \mid s_0 \sim \mu_0, \pi, P, t \sim \mathrm{Geo}(1-\gamma)\right) = \bar{\rho}_{\mu_0}^{\pi}(s),$$

$$\mathbb{P}\left(S_t = s, A_t = a \mid s_0 \sim \mu_0, \pi, P, t \sim \mathrm{Geo}(1-\gamma)\right) = \bar{\nu}_{\mu_0}^{\pi}(s, a).$$

$$\Longrightarrow \quad S_t \sim \bar{\rho}_{\mu_0}^{\pi} \quad \text{and} \quad (S_t, A_t) \sim \bar{\nu}_{\mu_0}^{\pi}.$$

# Collecting Labeled Data to Learn $Q^\pi$

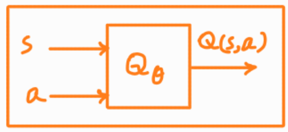



- For $i = 1, 2, \ldots, N$:
  - Sample a trajectory to roll in a state-action pair $(s^i, a^i)$ — feature $x^i$
  - Sample a trajectory to roll out from $(s^i, a^i)$ and obtain return $y^i$ — label $y^i$

$$\boxed{\begin{array}{c} (x^1 = (s^1, a^1), y^1) \\ (x^2 = (s^2, a^2), y^2) \\ \vdots \\ (x^N = (s^N, a^N), y^N) \end{array}}$$

## Supervised Learning of $Q^\pi$

Based on the data collected, we can form an estimate $\hat{Q}^\pi$ by solving the following problem:




$$\hat{Q}^\pi = \arg\min_{Q \in \mathcal{Q}} \sum_{i=1}^{N} (Q(s^i, a^i) - y^i)^2 .$$

Notice that without parameterization of $Q$, one can estimate it by

$$\hat{Q}^\pi(s,a) = \frac{1}{N} \sum_{i=1}^{N} y^i_{s,a} .$$

However, this approach fails in the case of parameterization, because it does not respect the structure imposed on $Q_\theta$. Also, such $\hat{Q}$ would not generalize to unseen state-action pairs. Hence, one needs to apply

$$\hat{Q}^\pi_\theta = \arg\min_{Q_\theta \in \mathcal{Q}} \underbrace{\frac{1}{N}}_{\text{does not matter}} \sum_{i=1}^{N} (Q_\theta(s^i, a^i) - y^i)^2 .$$

**Approximate Policy Iteration:** Similar to policy iteration, we apply policy evaluation and policy improvement iteratively;

$$\begin{tikzcd} \boxed{\text{policy } \pi} \arrow[r, bend left, "\text{policy evaluation}"] & \boxed{\text{value function } Q} \arrow[l, bend left, "\text{policy improvement}"] \end{tikzcd}$$

however, for policy evaluation, we use the supervised learning approach.

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Iterative method:

$\boxed{\text{On-policy method}}$

- Initialize $\pi_0 : \mathcal{S} \to \mathcal{A}$
- For $t = 0, 1, 2, \dots, T-1$ :

supervised learning for policy evaluation

- Evaluate $\pi_t$ by collecting a data set $\{(x^i = \overbrace{(s^i, a^i)}^{\text{roll in}}, \underbrace{y^i}_{\text{roll out}})\}_{i=1}^{N}$ and estimating $\hat{Q}^{\pi}$ using

$$\hat{Q}^{\pi_t} = \underset{Q \in \mathcal{Q}}{\arg\min} \sum_{i=1}^{N} \left(Q(s^i, a^i) - y^i\right)^2$$

policy improvement

- Improve policy by

$$\pi_{t+1}(s) = \underset{a \in \mathcal{A}}{\arg\max}\ \hat{Q}^{\pi_t}(s, a) \qquad \forall s \in \mathcal{S}$$

- Return $\pi_T$

**Question:** Does approximate policy iteration enjoy guaranteed monotonic improvement, i.e., does the following hold true?

$$V^{\pi_{t+1}}(s) \geq V^{\pi_t}(s) \quad \forall s \in S$$

**Permormance Difference Lemma:** Consider two policies $\pi : S \longrightarrow \Delta(A)$ and $\pi' : S \longrightarrow \Delta(A)$ over an infinite-horizon discounted MDP.

The value functions of policies $\pi$ and $\pi'$ over the MDP satisfy the following:

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \mathbb{E}_{a \sim \pi(\cdot|s)} \left[ Q^{\pi'}(s,a) - V^{\pi'}(s) \right] \right].$$

**Proof:**

definition of value function

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \mathbb{E}_{\substack{A_t \sim \pi \\ S_{t+1} \sim P(\cdot|S_t, A_t)}} \left[ \sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \,\middle|\, S_0 = s \right] - V^{\pi'}(s_0)$$

Bellman consistency equation + rearranging terms

$$= \mathbb{E}_{A_0 \sim \pi} \left[ R(s_0, A_0) + \gamma \mathbb{E}_{S_1 \sim P} \left[ V^{\pi}(S_1) \right] - V^{\pi'}(s_0) \right]$$

add and subtract

$$= \mathbb{E}_{A_0 \sim \pi} \left[ R(s_0, A_0) + \gamma \mathbb{E}_{S_1 \sim P} \left[ V^{\pi}(S_1) - V^{\pi'}(S_1) + V^{\pi'}(S_1) \right] - V^{\pi'}(s_0) \right]$$

Bellman consistency equation + rearranging terms

$$= \gamma \, \mathbb{E}_{\substack{A_0 \sim \pi \\ S_1 \sim P}} \left[ \underline{V^{\pi}(S_1) - V^{\pi'}(S_1)} \right] + \mathbb{E}_{A_0 \sim \pi} \left[ Q^{\pi'}(S_0, A_0) - V^{\pi'}(S_0) \right]$$

repeat the argument for $V^{\pi}(S_1) - V^{\pi'}(S_1)$

$$= \gamma^2 \, \mathbb{E}_{\substack{A_0, A_1 \sim \pi \\ S_1, S_2 \sim P}} \left[ V^{\pi}(S_2) - V^{\pi'}(S_2) \right]$$

$$+ \gamma \, \mathbb{E}_{\substack{A_0, A_1 \sim \pi \\ S_1 \sim P}} \left[ Q^{\pi'}(S_1, A_1) - V^{\pi'}(S_1) \right]$$

$$+ \mathbb{E}_{A_0 \sim \pi} \left[ Q^{\pi'}(S_0, A_0) - V^{\pi'}(S_0) \right]$$

$$\vdots$$

after K iterations

$$= \overbrace{\gamma^{K} \, \mathbb{E}_{\substack{A_0, A_1, \dots, A_{K-1} \sim \pi \\ S_1, S_2, \dots, S_K \sim P}} \left[ V^{\pi}(S_K) - V^{\pi'}(S_K) \right]}^{\text{goes to zero as } K \to \infty}$$

$$+ \sum_{i=0}^{K-1} \gamma^{i} \, \mathbb{E}_{\substack{A_0, A_1, \dots, A_i \sim \pi \\ S_1, S_2, \dots, S_i \sim P}} \left[ Q^{\pi'}(S_i, A_i) - V^{\pi'}(S_i) \right]$$

$K \to \infty$ + definition of expectation + definition of occupancy measure

$$= \frac{1}{1-\gamma} \, \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \mathbb{E}_{a \sim \pi(\cdot|s)} \left[ Q^{\pi'}(s, a) - V^{\pi'}(s) \right] \right]$$

$\blacksquare$

**Advantage function:** The advantage function of a policy $\pi$, $A^\pi : S \times A \to \mathbb{R}$, is defined as

$$\underbrace{A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)}_{\text{the advantage of taking action } a \text{ instead of following policy } \pi \text{ at state } s} \qquad \forall s \in S,\ \forall a \in A.$$

* For a deterministic policy $\pi$, it holds that

$$V^\pi(s) = \mathbb{E}_{a\sim\pi(s)}\left[Q^\pi(s,a)\right] = Q^\pi(s,\pi(s))$$

$$A^\pi(s,\pi(s)) = Q^\pi(s,\pi(s)) - V^\pi(s) = 0$$

* In the policy improvement step of PI, we apply

$$\pi_{t+1}(s) = \arg\max_{a\in A} Q^{\pi_t}(s,a)$$

$$= \arg\max_{a\in A} \left(Q^{\pi_t}(s,a) - V^{\pi_t}(s)\right) = \underbrace{\arg\max_{a\in A} A^{\pi_t}(s,a)}_{\text{action with largest advantage}}.$$

* The performance difference lemma can be rewritten as

$$V^\pi(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim\bar{\rho}^\pi_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}\left[A^{\pi'}(s,a)\right]\right]$$

**Corollary:** In the setting considered in the performance difference lemma, it holds that

$$|V^{\pi}(s_0) - V^{\pi'}(s_0)| \leq \frac{R_{max}}{(1-\gamma)^2} \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \| \pi(\cdot|s) - \pi'(\cdot|s) \|_1 \right],$$

where $R_{max} = \max_{s,a} |R(s,a)|$.

**Proof:** Note that

$$\mathbb{E}_{a \sim \pi(\cdot|s)} \left[ Q^{\pi'}(s,a) - V^{\pi'}(s) \right]$$

$V^{\pi'}$ independent of $a$ ←
$$= \mathbb{E}_{a \sim \pi(\cdot|s)} \left[ Q^{\pi'}(s,a) \right] - V^{\pi'}(s)$$

definition of expectation + Bellman consistency equation ←
$$= \sum_{a \in \mathcal{A}} \pi(a|s) Q^{\pi'}(s,a) - \sum_{a \in \mathcal{A}} \pi'(a|s) Q^{\pi'}(s,a)$$

$$= \sum_{a \in \mathcal{A}} (\pi(a|s) - \pi'(a|s)) Q^{\pi'}(s,a).$$

$$|V^{\pi}(s_0) - V^{\pi'}(s_0)| = \left| \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \sum_{a \in \mathcal{A}} (\pi(a|s) - \pi'(a|s)) Q^{\pi'}(s,a) \right] \right|$$

triangle inequality ←
$$\leq \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \sum_{a \in \mathcal{A}} |\pi(a|s) - \pi'(a|s)| \underbrace{|Q^{\pi'}(s,a)|}_{\leq \frac{R_{max}}{1-\gamma}} \right]$$

definition of L1 norm ←
$$\leq \frac{R_{max}}{(1-\gamma)^2} \mathbb{E}_{s \sim \bar{\rho}^{\pi}_{s_0}} \left[ \| \pi(\cdot|s) - \pi'(\cdot|s) \|_1 \right]$$

∎

**Policy improvement in policy iteration:** In policy iteration, we apply

$$\pi_{t+1}(s) = \arg\max_{a \in A} A^{\pi_t}(s,a).$$

We can show that

$$V^{\pi_{t+1}}(s_0) - V^{\pi_t}(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} \left[ \mathbb{E}_{a \sim \pi_{t+1}(\cdot|s)} \left[ A^{\pi_t}(s,a) \right] \right]$$

$$\text{determinism of } \pi_{t+1} \leftarrow = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} \left[ A^{\pi_t}(s, \pi_{t+1}(s)) \right]$$

$$\text{definition of } \pi_{t+1} \leftarrow \geq \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} \left[ A^{\pi_t}(s, \pi_t(s)) \right]$$

$$\text{definition of } A^{\pi_t} \leftarrow = 0$$

$$\Longrightarrow V^{\pi_{t+1}}(s_0) \geq V^{\pi_t}(s_0). \quad \boxed{\text{monotonic improvement}}$$

**Note:** Different from policy iteration, approximate policy iteration does not guarantee monotonic policy improvement.

This is due to the fact the the occupancy measures of the policies from two consecutive iterations can differ significantly.